

# Daviti Gochitashvili

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## EDUCATION

**Binghamton University, State University of New York**

**Vestal, NY, USA**

*PhD in Computational Physics* | GPA: 4.00

**May 2026**

*Master of Science in Physics* | GPA: 4.00

**August 2022**

Provost's Summer Fellowship - *Competitive four-year research funding awarded to select doctoral candidates*

**Free University of Tbilisi**

**Tbilisi, Georgia**

*Bachelor of Science in Physics* | GPA: 3.92

**August 2019**

Full-Tuition Scholarship - *Competitively awarded to the top 10 students in the national physics Olympiad; placed 9th*

## SKILLS & CERTIFICATIONS

- **Programming & Tools:** Python, C/C++, Java, SQL, PL/SQL, Bash, Slurm (HPC), Git, Linux/Unix, Excel
- **Python Libraries:** NumPy, SymPy, Pandas, SciPy, Matplotlib, Seaborn, scikit-learn, yfinance
- **Quantitative:** Options pricing models, Portfolio construction & backtesting, Time-series, Stochastic calculus, Optimization
- **Languages:** Georgian (native), English (fluent), Russian (conversational), German (basic)
- **Certifications:** The Erdős Institute Quant Finance Boot Camp ([2025](#)), Wharton School UPenn - Finance & Quantitative Modeling for Analysts ([Coursera, 2025](#)), Construct Stock Market Indices ([Coursera, 2025](#)), SQL ([HackerRank, 2020](#))

## PROJECTS

**Options Pricing, Independent Project**

**September 2025**

- Built a Python library covering fundamentals (Black-Scholes, Greeks), numerical methods (Monte Carlo, PDE solvers), advanced models (Heston, American options), and backtesting: includes 117 unit tests and 21 notebook demos ([link](#))

**Portfolio Construction, Independent Project**

**October 2025**

- Developed a Python pipeline to retrieve historical market data, construct and benchmark a free-float-weighted equity index, and generate correlation matrices, heatmaps, and performance visualizations ([link](#))

## FINANCIAL INDUSTRY EXPERIENCE

**Bank of Georgia, Analyst - Developer** (Tbilisi, Georgia)

**July 2019 - August 2020**

- Developed and deployed API services for the core banking system using **Java** and **PL/SQL**, improving responsiveness, streamlining workflows, and supporting high-volume transaction processing
- Designed and implemented 18 banking and quantitative modules for the Cash Operations and Liens **Agile (Scrum)** development squad, enabling faster delivery and more data-driven decision-making
- Collaborated with 7 Revenue Service officials to gather requirements, interpret complex banking rules, ensure compliance, and translate them into automated system functions
- Improved cash-counting operations by about 300% by building **Java**-based automation and reporting workflows around money-counting machines, using complex **SQL** queries for analytics and process validation

## ACADEMIC EXPERIENCE

**Binghamton University, Graduate Researcher / Associate** (Vestal, NY, USA)

**September 2020 - May 2026**

- Contributed to the development of a **C/Python** package for building, testing, and deploying **machine learning** models for atomic interactions through integrated **Slurm/HPC** workflows, increasing overall model accuracy by 20%
- Utilized neural-network interatomic potentials and evolutionary algorithms to discover new stable compounds, screening over 1 million candidate phases and identifying 29 novel materials on the thermodynamic convex hull
- Implemented high-dimensional optimization methods ML-based potentials to navigate complex energy surfaces more efficiently, improving global-minimum discovery rates in statistical benchmarks across more than 2 million structures

**Binghamton University, Teaching Assistant** (Vestal, NY, USA)

**August 2020 - May 2022**

- Instructed general/quantum physics discussion sections of 30 students each and provided individual student support
- Conducted lab demonstrations, set up experiments, and assisted students across 4 sections during the sessions

## PUBLICATIONS (29 citations)

- Thorn, A., **Gochitashvili, D.**, Kharabadze, S., Kolmogorov, A. N. Machine learning search for stable binary Sn alloys with Na, Ca, Cu, Pd, and Ag. in *Phys. Chem. Chem. Phys.* ([2023](#))
- Tomassetti, C. R., **Gochitashvili, D.**, Renskers, C., Margine, E. R., Kolmogorov, A. N. First-principles design of ambient-pressure  $\text{Mg}_x\text{B}_2\text{C}_2$  and  $\text{Na}_x\text{BC}$  superconductors. in *Phys. Rev. Mater.* ([2024](#))
- **Gochitashvili, D.**, Tomassetti, C. R., Margine, E. R., Kolmogorov, A. N. High-Tc  $\text{Ag}_x\text{BC}$  and  $\text{Cu}_x\text{BC}$  superconductors accessible via topochemical reactions. in *J. Mater. Chem. C.* ([2025](#))
- **Gochitashvili, D.**, Meyers, M., Wang, C., Kolmogorov, A. N. Improving structure search with hyperspatial optimization and TETRIS seeding. in *Phys. Chem. Chem. Phys.* ([2025](#))